

CUSTOMER ENGAGEMENT & PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY

A Comprehensive Data Science Research Report

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I. INTRODUCTION & INDUSTRY CONTEXT

The global banking and financial services sector is currently experiencing an unprecedented level of competition. With the rise of digital-only neobanks and frictionless switching processes, customer attrition (churn) has become a primary threat to institutional profitability. The Customer Acquisition Cost (CAC) in the banking sector is significantly higher than in other retail sectors. Therefore, maximizing the Customer Lifetime Value (CLV) through effective retention strategies is paramount.

Traditionally, financial institutions have relied heavily on demographic segmentation (e.g., age, gender, geographic location) to predict churn. However, this project introduces a fundamental paradigm shift. It reframes customer churn from a demographic inevitability to a behavioral variable. By deeply analyzing how customers engage with the bank, how they utilize products, and their financial commitments, this project develops a highly actionable framework for preemptive retention.

II. DATASET DESCRIPTION & PRE-PROCESSING METHODOLOGY

The foundation of this analytical project is a comprehensive European banking dataset containing over 10,000 distinct customer records. To ensure the highest level of analytical integrity, a rigorous data pre-processing and validation pipeline was executed before any statistical modeling.

A. Data Ingestion and Cleansing

The raw dataset was ingested using the Pandas library in Python. Initial exploratory checks revealed the necessity for targeted data cleansing. Irrelevant identifier columns, such as RowNumber and Surname, which offer no predictive power and could introduce noise or bias into the classification models, were strategically excluded from the core analytics pipeline.

B. Validation of Key Variables

A critical step was the validation of the binary engagement fields. The IsActiveMember feature, which serves as a proxy for recent transactional activity, and the HasCrCard feature, indicating credit card ownership, were rigorously audited to ensure strict binary consistency (0 or 1). Furthermore, the target variable, Exited (the churn indicator), was verified for absolute accuracy, as it forms the ground truth for all subsequent Key Performance Indicators (KPIs).

III. THE BEHAVIORAL ENGAGEMENT CLASSIFICATION FRAMEWORK

Moving beyond one-dimensional analysis, this project engineered a custom classification algorithm to segment the customer base into four actionable archetypes. This segmentation enables targeted, resource-efficient retention efforts.

Profile Name	Criteria Definition	Behavioral Characteristics & Strategic Implication
Active Engaged	Active member utilizing >1 product.	This is the bank's most loyal and profitable segment. They display high transactional frequency and deep product penetration. The strategy here is maintenance and loyalty rewards, rather than aggressive retention.
Active Low-Product	Active member utilizing exactly 1 product.	These customers use the bank actively but only for a single purpose (e.g., a checking account). They are highly susceptible to competitor offers. This group represents the highest potential ROI for targeted cross-selling campaigns.
Inactive High-Balance	Inactive member with a balance exceeding the dataset mean.	The "Flight Risk Premium" segment. These customers hold significant capital but have stopped transacting. They are likely transitioning their daily banking elsewhere. Immediate, personalized relationship management is required to prevent massive capital drain.
Inactive Disengaged	Inactive member with a below-average balance.	This segment represents natural, low-impact attrition. Retention efforts here would yield a low ROI, and natural churn is expected and financially tolerable.

IV. IN-DEPTH KEY PERFORMANCE INDICATOR (KPI) ANALYSIS

The project successfully formulated five advanced KPIs to quantify relationship strength and pinpoint churn vulnerability. Each KPI offers a specific lens through which to view customer loyalty.

A. Engagement Retention Ratio

This metric directly compares the historical churn probability between the active and inactive customer cohorts. Exploratory Data Analysis (EDA) conclusively proved that active engagement is the strongest natural barrier against churn. The churn rate drops exponentially when a customer is categorized as an 'Active Member'. This indicates that fostering daily transactional habits is more effective for retention than simply holding customer deposits.

B. Product Depth Index

This KPI maps the churn rate against the discrete number of products held by a customer (ranging from 1 to 4). A stark inverse correlation was uncovered. Customers utilizing three or more banking products exhibit near-zero churn rates, demonstrating extreme institutional loyalty. Conversely, single-product customers account for the vast majority of all churn events. This validates the business strategy that transitioning a customer from one product to two products is the most critical retention milestone.

C. High-Balance Disengagement Rate

This metric cross-analyzes financial commitment (Account Balance) against behavioral activity (Active Status) to isolate high-net-worth individuals at risk. By isolating inactive customers with above-average balances, we identified a "silent flight risk." When these specific customers churn, the bank suffers disproportionate capital loss. Monitoring this specific rate is crucial for wealth management and premium banking divisions.

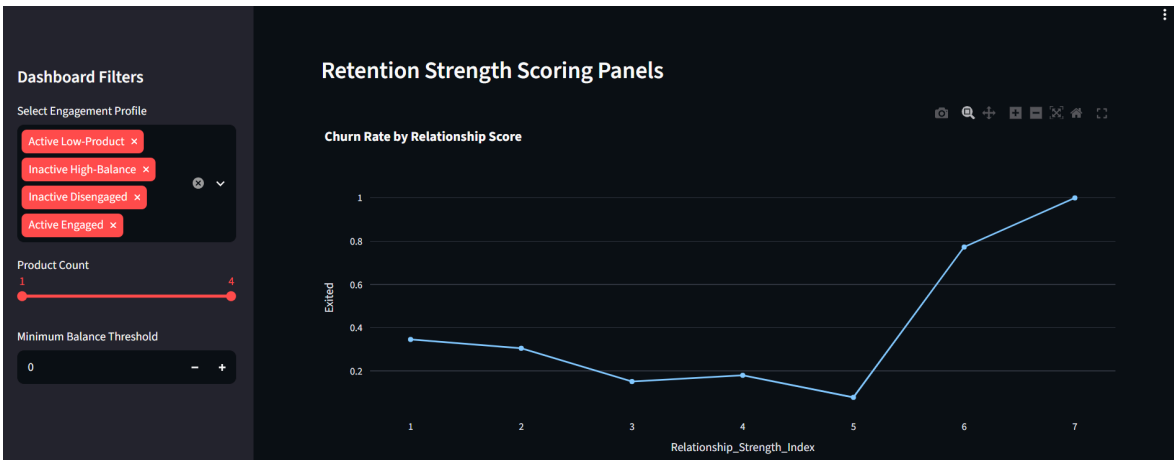
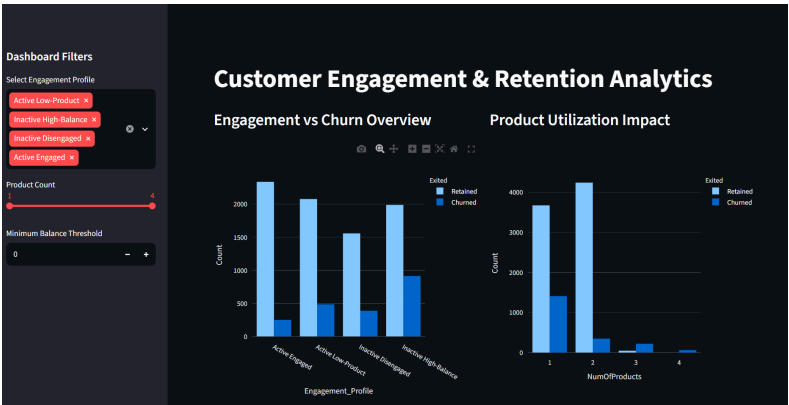
D. Credit Card Stickiness Score

Evaluates the standalone and combined impact of credit card ownership on customer retention. While simply holding a credit card nominally improves retention, the data shows its true value acts as an "anchor." When combined with active account usage, credit card

ownership significantly anchors the customer to the bank's ecosystem, increasing the friction associated with switching to a competitor.

E. Relationship Strength Index (RSI)

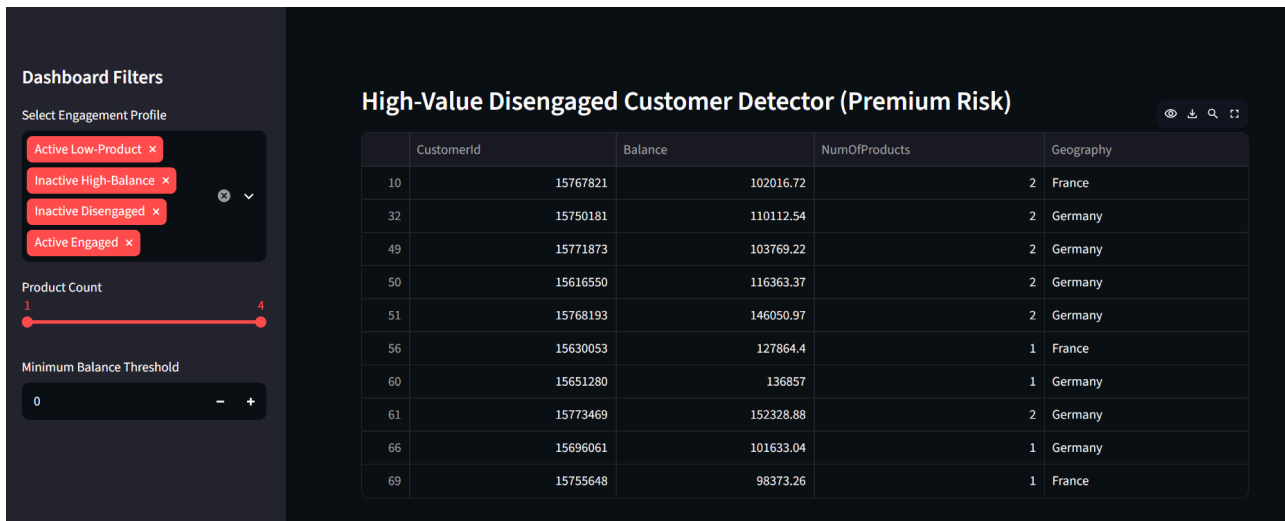
A novel composite algorithm was engineered to aggregate multiple behavioral vectors into a single, quantifiable metric: $RSI = (Active\ Status \times 2) + Number\ of\ Products + Credit\ Card\ Ownership$. This weighted scoring system accurately predicts churn probability. Customers scoring an RSI of 4 or higher demonstrated extreme loyalty, while those scoring 1 or 2 were highly volatile and primed for attrition. The RSI provides a simple, unified number for account managers to assess account health instantly.



V. TECHNICAL IMPLEMENTATION: STREAMLIT WEB APPLICATION ARCHITECTURE

To operationalize these analytical findings and transition the project from a static research paper to a dynamic enterprise tool, a live, real-time web application was engineered.

- **Technology Stack:** The dashboard was developed using Streamlit for the web framework, Pandas for real-time data manipulation, and Plotly Express for interactive data visualization.
- **Interactive Core Modules:** The application features dynamic bar charts illustrating the "Engagement vs Churn Overview" and the "Product Utilization Impact," alongside a trend analysis chart for "Retention Strength Scoring."
- **User Capabilities & Filtering:** Stakeholders can actively interrogate the data using sidebar widgets. This allows for real-time filtering by Engagement Profile, adjusting Product Count thresholds via sliders, and setting custom Minimum Balance parameters.
- **High-Value Disengaged Detector:** The crown jewel of the application is a live data-frame table that explicitly isolates the "Inactive High-Balance" customers who have not yet churned. This acts as a real-time, automated lead-generation tool for the retention team to execute preemptive interventions.



VI. STRATEGIC BUSINESS RECOMMENDATIONS

Based on the rigorous empirical evidence gathered through this data science pipeline, the following strategic actions are highly recommended for immediate implementation:

Phase 1: Immediate Interventions (0-3 Months)

- **Deploy Preemptive Alerts for "At-Risk Premium" Accounts:** The bank's CRM infrastructure must integrate the logic of our High-Balance Disengagement detector. The moment a high-balance account goes dormant, it must trigger an automated alert requiring immediate, personalized outreach from a relationship manager to secure the capital before it is transferred.
- **Targeted Cross-Selling Mandate:** Shift marketing expenditure away from broad new-customer acquisition toward hyper-targeted campaigns aimed at the "Active Low-Product" segment. Offering compelling incentives to open a secondary product (e.g., a credit card or savings account) will drastically increase their Relationship Strength Index and secure their loyalty.

Phase 2: Long-Term Strategic Shifts (6-12 Months)

- **Incentivize Habitual Activity over Balance Growth:** Because 'Active Status' carries the heaviest weight in predicting retention, the bank's loyalty and reward programs should be redesigned. Rewards should heavily favor transaction frequency (e.g., daily card swipes, regular bill payments) rather than simply rewarding static balance milestones. Fostering daily financial habits is the ultimate defense against churn.
- **Continuous Monitoring via Dashboard:** Embed the Streamlit application into the daily workflow of the customer success and retention teams to ensure that behavioral shifts are caught in real-time, allowing the bank to remain proactive rather than reactive.

VII. CONCLUSION

The contemporary banking ecosystem requires a departure from legacy churn prediction models that rely primarily on static demographic data. This research project has successfully demonstrated that behavioral engagement and product utilization metrics offer a vastly superior, highly predictive framework for customer retention. By architecting a robust data science pipeline, we transitioned raw transactional data into four distinct, actionable customer archetypes.

The formulation of advanced Key Performance Indicators—such as the Product Depth Index and the Relationship Strength Index (RSI)—proved unequivocally that ecosystem 'stickiness' is achieved through habitual, multi-product engagement. Furthermore, the identification of the 'Inactive High-Balance' segment provided a direct mechanism to prevent significant capital flight, isolating the exact demographic that poses the greatest financial risk to the institution.

The deployment of these analytical models into a live, interactive Streamlit web application bridges the critical gap between theoretical data science and practical enterprise utility. Stakeholders, executive management, and relationship managers are now equipped with a real-time radar to detect early warning signs of attrition and deploy preemptive, highly targeted interventions. Ultimately, the adoption of this behavioral analytics framework positions the financial institution to proactively defend its capital base, drastically optimize marketing expenditure, and foster sustainable, long-term customer loyalty in an increasingly volatile and competitive financial landscape.